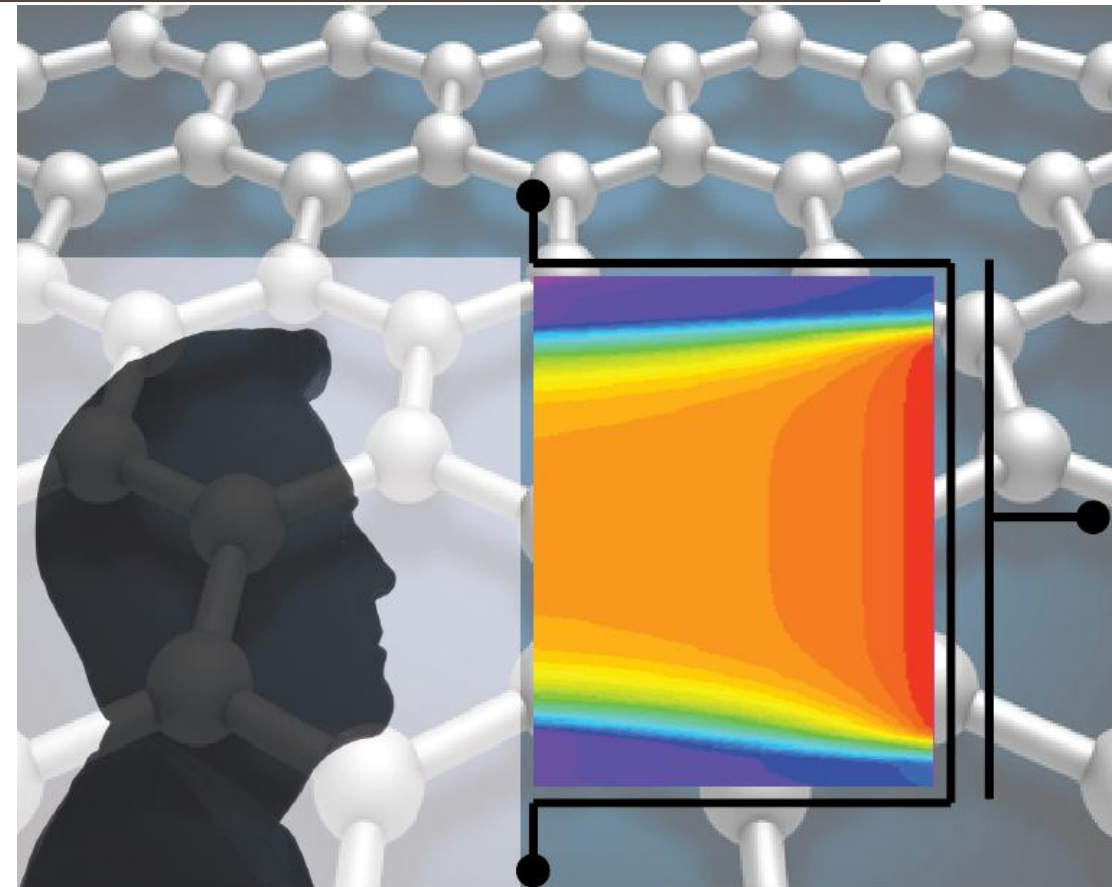


ADVANCED SOLID-STATE DEVICES

Week-9, Lecture-2

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20th March, 2025



MOSFET



Frequency Limitations

MOSFET: Frequency Response

- Transit time of carriers through the channel is usually not the limiting factor in the frequency response of MOSFETs
- The time required in charging/discharging of the parasitic capacitors is the limiting factor
- Let us understand the frequency limitations using a simplified high-frequency small-signal equivalent circuit

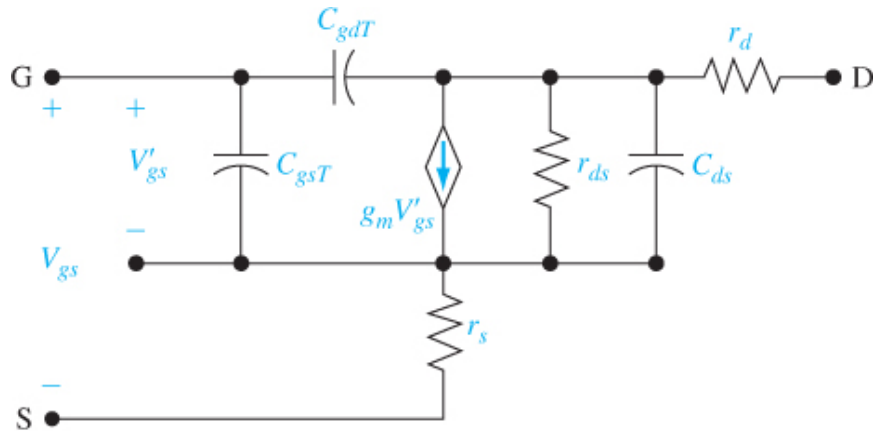


Figure 10.53 | Small-signal equivalent circuit of a common-source n-channel MOSFET.

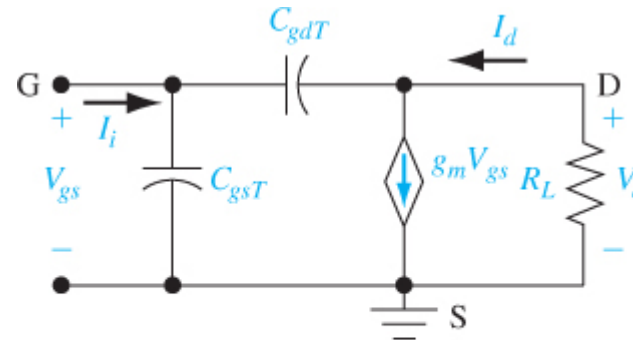


Figure 10.56 | High-frequency small-signal equivalent circuit of common-source n-channel MOSFET.

- We have taken gate capacitance, current source, and load resistor R_L
- We have neglected r_s , r_d , r_{ds} , and C_{ds} to get an approximate result

Frequency Response

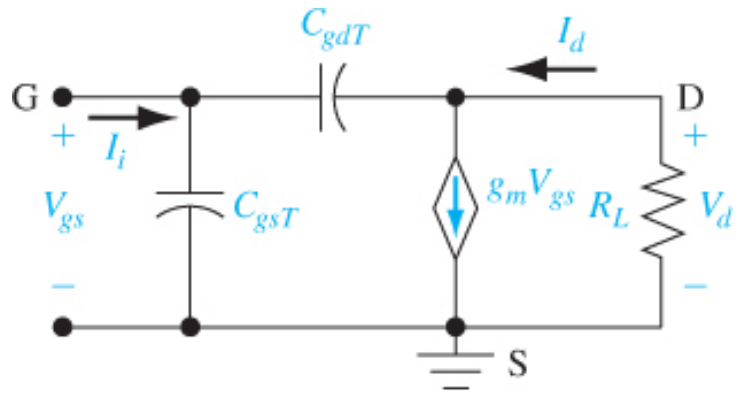


Figure 10.56 | High-frequency small-signal equivalent circuit of common-source n-channel MOSFET.

Problem: Formulate Kirchhoff's Current Law equations at the input node and output node in terms I_i , V_{gs} , V_d and circuit components [Assume angular frequency of the input signal is ω]

Solution:

$$I_i = j\omega C_{gsT} V_{gs} + j\omega C_{gdT} (V_{gs} - V_d)$$

$$g_m V_{gs} + \frac{V_d}{R_L} = j\omega C_{gdT} (V_{gs} - V_d)$$

Eliminating V_d (will require some trivial algebraic manipulations):

$$I_i = j\omega \left[C_{gsT} + C_{gdT} \left(\frac{1 + g_m R_L}{1 + j\omega R_L C_{gdT}} \right) \right] V_{gs}$$

Normally, $j\omega R_L C_{gdT} \ll 1$

Hence, $I_i = j\omega [C_{gsT} + (1 + g_m R_L) C_{gdT}] V_{gs}$

Frequency Response: Impact of Miller Capacitance

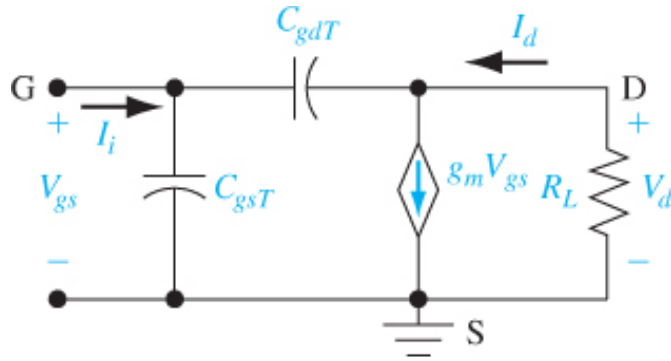


Figure 10.56 | High-frequency small-signal equivalent circuit of common-source n-channel MOSFET.

$$I_i = j\omega [C_{gsT} + (1 + g_m R_L) C_{gdT}] V_{gs}$$

- We can replace C_{gdT} with an equivalent capacitance to the ground $C_M = (1 + g_m R_L) C_{gdT}$
- C_M is referred to as Miller capacitance

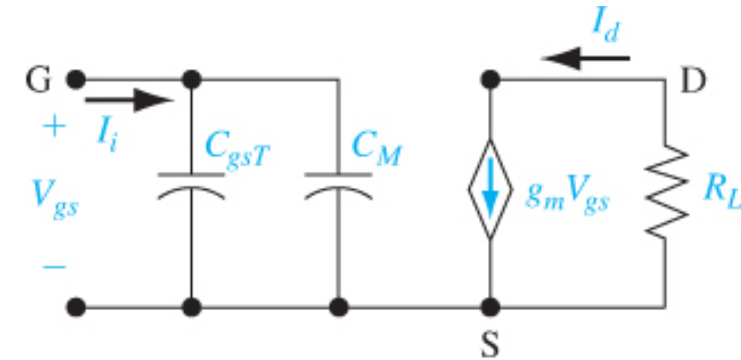


Figure 10.57 | Small-signal equivalent circuit including Miller capacitance.

- The gate-drain parasitic capacitance is multiplied by $(1 + g_m R_L)$
- Hence, despite being small, C_{gdT} can lead to significant input capacitance

Frequency Response: Current Gain

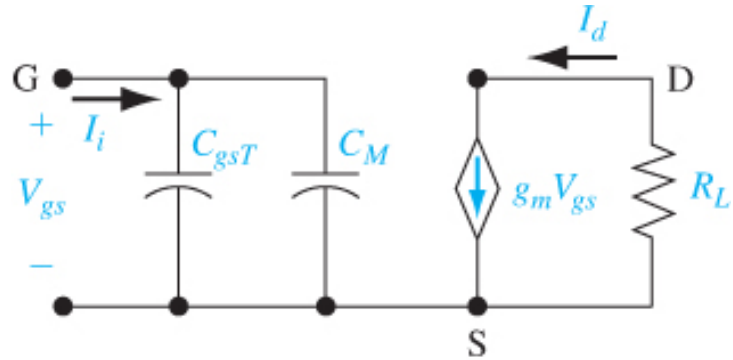


Figure 10.57 | Small-signal equivalent circuit including Miller capacitance.

$$I_i = j\omega [C_{gsT} + C_M] V_{gs}$$

$$I_d = g_m V_{gs}$$

Current gain magnitude $\left| \frac{I_d}{I_i} \right|$

$$= \frac{g_m}{\omega [C_{gsT} + C_M]}$$

$$= \frac{g_m}{2\pi f [C_{gsT} + C_M]}$$

Frequency Response: Cutoff Frequency (1)

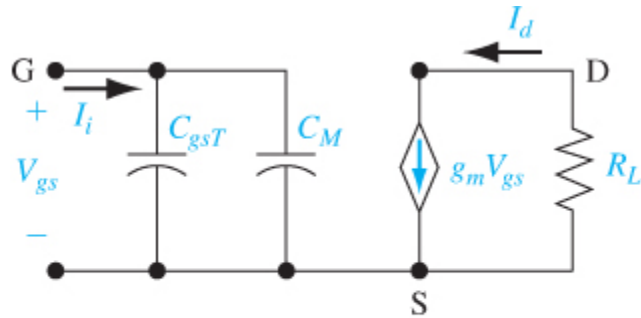


Figure 10.57 | Small-signal equivalent circuit including Miller capacitance.

$$\text{Current gain } \left| \frac{I_d}{I_i} \right| = \frac{g_m}{2\pi f [C_{gsT} + C_M]}$$

- Cutoff frequency f_T is the frequency at which the current gain becomes 1

$$\text{Hence, } \frac{g_m}{2\pi f_T [C_{gsT} + C_M]} = 1$$
$$f_T = \frac{g_m}{2\pi [C_{gsT} + C_M]}$$

Frequency Response: Cutoff Frequency (2)

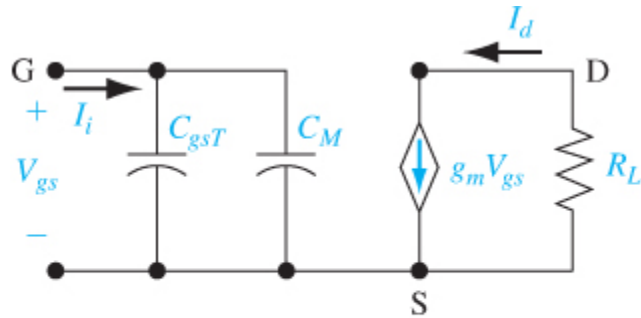


Figure 10.57 | Small-signal equivalent circuit including Miller capacitance.

$$f_T = \frac{g_m}{2\pi[C_{gsT} + C_M]}$$

- For an ideal case:
 - $C_M = 0$ [In saturation gate-drain capacitance is zero and if gate-drain overlap is negligible $C_{gdT} = 0$]
 - $C_{gsT} = WLC'_{ox}$ [oxide capacitance]
 - $g_m = \frac{\mu_n W C'_{ox}}{L} (V_{GS} - V_T)$ [We have computed earlier]
- Hence, $f_T = \frac{g_m}{2\pi W L C'_{ox}} = \frac{\mu_n W C'_{ox}}{2\pi W L C'_{ox} L} (V_{GS} - V_T)$

$$= \frac{\mu_n}{2\pi L^2} (V_{GS} - V_T)$$

We can increase the cut-off frequency by:

- Using material of higher mobility
- Reducing channel length
- Increasing gate overdrive

MOSFET: Problem

Problem: Consider an ideal n-channel MOSFET with the following parameters:

$$L = 4 \mu m$$

$$\mu_n = 400 \text{ cm}^2/Vs$$

$$V_{GS} = 3 \text{ V}$$

$$V_T = 1 \text{ V}$$

Calculate the cutoff frequency. Assume that the mobility is constant.

$$f_T = \frac{\mu_n}{2\pi L^2} (V_{GS} - V_T)$$

Answer:

$$f_T = 796 \text{ MHz}$$

MOSFET: Non-Ideal Effects



Channel Length Modulation

Channel Length Modulation: Basics

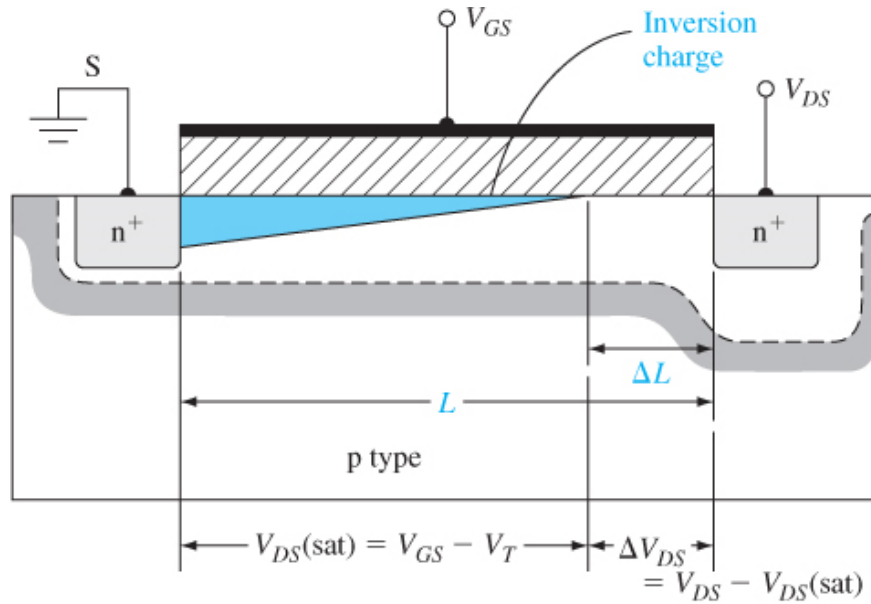


Figure 11.5 | Cross-section of an n-channel MOSFET showing the channel length modulation effect.

- Ideal MOSFET: channel length L is constant
- The depletion region near the drain (i.e., pinch of point) extends into the channel when V_{DS} is increased beyond saturation
- Effective channel length reduces
- Hence, channel length depends on the applied bias V_{DS} in the saturation region
- This effect is known as *channel length modulation*

- **Observation of channel length modulation:** drain current increases in the output characteristics in saturation region (in contrast to ideal MOSFET in which drain current do not change with V_{DS})
- Channel length modulation becomes more prominent in MOSFETs with smaller channel lengths

Channel Length Modulation: Computation

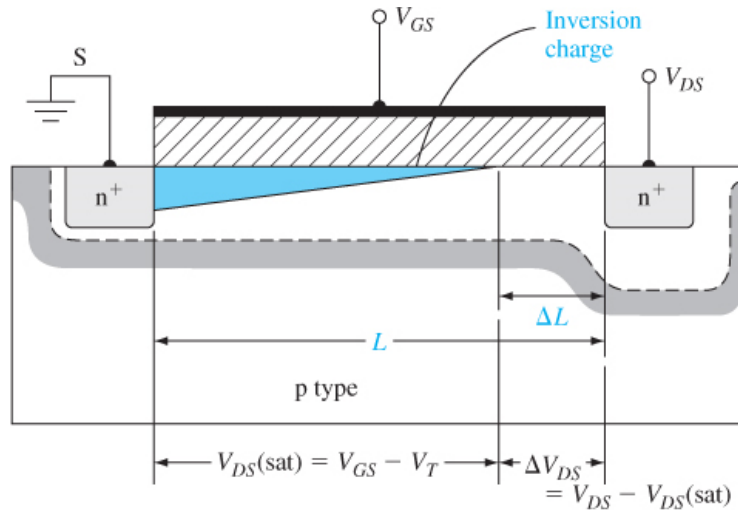


Figure 11.5 | Cross-section of an n-channel MOSFET showing the channel length modulation effect.

- Under zero drain bias, depletion region width extending into p-substrate (considering n⁺p junction) is:

$$x_p = \sqrt{\frac{2\epsilon_s \phi_{fp}}{qN_a}}$$

- Due to reverse bias V_{DS} to the n⁺p junction, the depletion region extends into p-substrate:

$$x_p = \sqrt{\frac{2\epsilon_s}{qN_a} (\phi_{fp} + V_{DS})}$$

- The depletion region (pinch point) starts extending into the channel only after saturation is reached
 - Decrease in channel length ΔL is zero until $V_{DS,sat}$
- Hence, we can write ΔL as (assuming $V_{DS} > V_{DS,sat}$):

$$\Delta L = \sqrt{\frac{2\epsilon_s}{qN_a} (\phi_{fp} + V_{DS})} - \sqrt{\frac{2\epsilon_s}{qN_a} (\phi_{fp} + V_{DS,sat})}$$

Channel Length Modulation Parameter

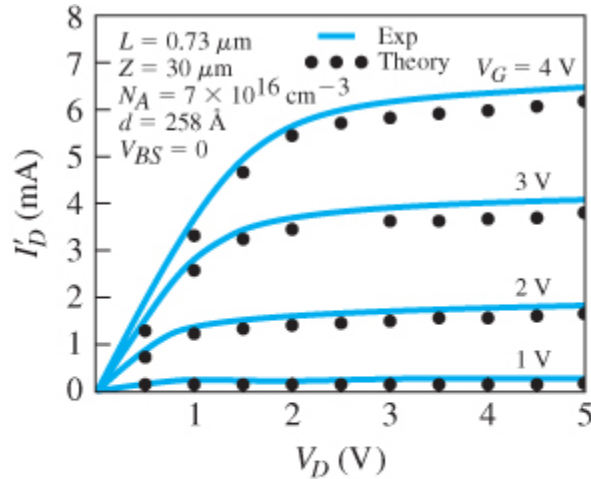


Figure 11.7 | Current-voltage characteristics of a MOSFET showing short-channel effects.
(From Sze [22].)

- Ideal MOSFET in saturation: $I_D = \frac{\mu_n W C'_{ox}}{2L} [V_{GS} - V_T]^2$

- Due to channel length modulation L becomes $L - \Delta L$

- So current becomes: $I'_D = \frac{L}{L - \Delta L} I_D$

- I'_D now becomes a function of V_{DS} because ΔL is a function of V_{DS}

- We can also write saturation current as:

$$I'_D = I_D (1 + \lambda V_{DS})$$

where λ is the channel length modulation parameter

- Drain conductance now becomes $g_d = \left. \frac{\partial I_D}{\partial V_{DS}} \right|_{V_{GS}=\text{constant}} = \lambda I_D$

- Note that for an ideal MOSFET, we had computed $g_d = 0$

MOSFET: Non-Ideal Effects



Velocity Saturation

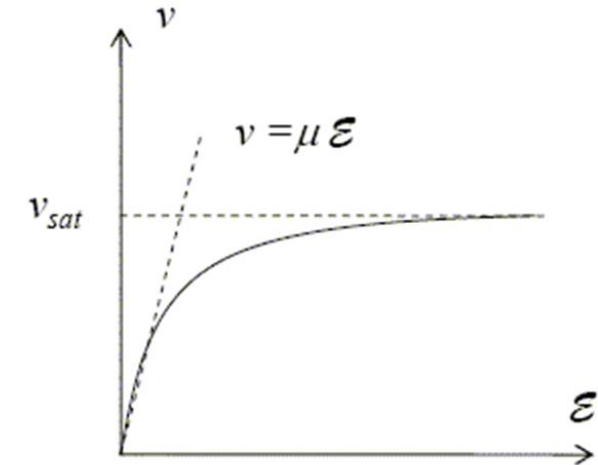
Velocity Saturation: Basics

- Ideal MOSFET in saturation: $I_D = \frac{\mu_n W C'_{ox}}{2L} [V_{GS} - V_T]^2$

- We had assumed that the mobility is a constant parameter
- This assumption implies that carrier drift velocity goes on increasing with increase in electric field: $v_d = \mu E$

- However, we have seen that carrier velocity saturates at high electric field
- When horizontal electric field is greater than $\sim 10^4$ V/cm velocity saturation occurs in silicon

- Velocity saturates to approximately $v_{d,sat} \sim 10^7$ cm/s
- Since $v_d = \mu E$, effectively mobility decreases at high electric field



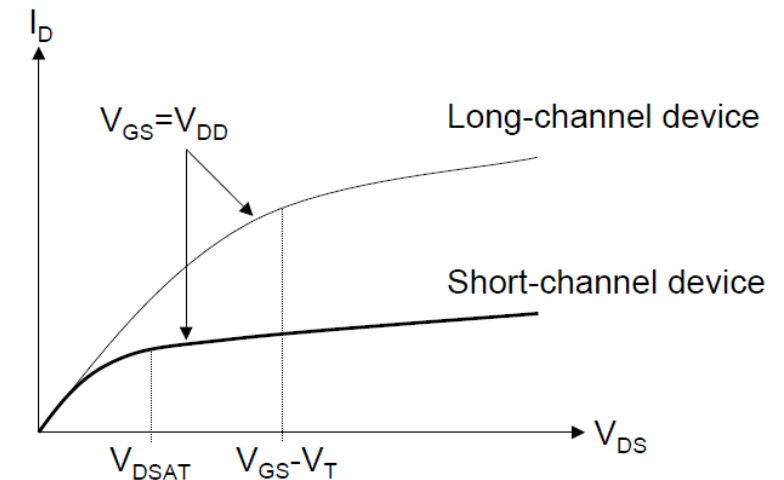
Velocity Saturation: Advanced Nodes

Problem: Consider $V_{DS} = 1$ V and channel length $L = 100$ nm. Compute the electric field along the channel.

Solution: $E = \frac{V}{L} = \frac{1}{100 \times 10^{-7}} = 10^5$ V/cm

⇒ Velocity Saturation occurs in MOSFET for $L < 100$ nm (advanced nodes or short-channel devices)

- Saturation occurs in an ideal MOSFET due to pinch-off when: $V_{DS} > (V_{DS,sat} = V_{GS} - V_T)$ for NMOS



- For short-channel MOSFET saturation occurs at lower drain voltage due to velocity saturation

Velocity Saturation: Effects

- Ideal MOSFET in saturation: $I_D = \frac{\mu_n W C'_{ox}}{2L} [V_{GS} - V_T]^2$
- Due to velocity saturation: $I_D = W C'_{ox} (V_{GS} - V_T - \frac{V_{DSAT}}{2}) v_{d,sat}$
- Effects of velocity saturation:
 - Saturation current varies linearly with V_{GS} rather than quadratic with V_{GS}
 - Saturation current becomes independent of channel length
- The transconductance and cutoff frequency will also be impacted by velocity saturation (check yourself by putting the expressions of current in the respective equations)

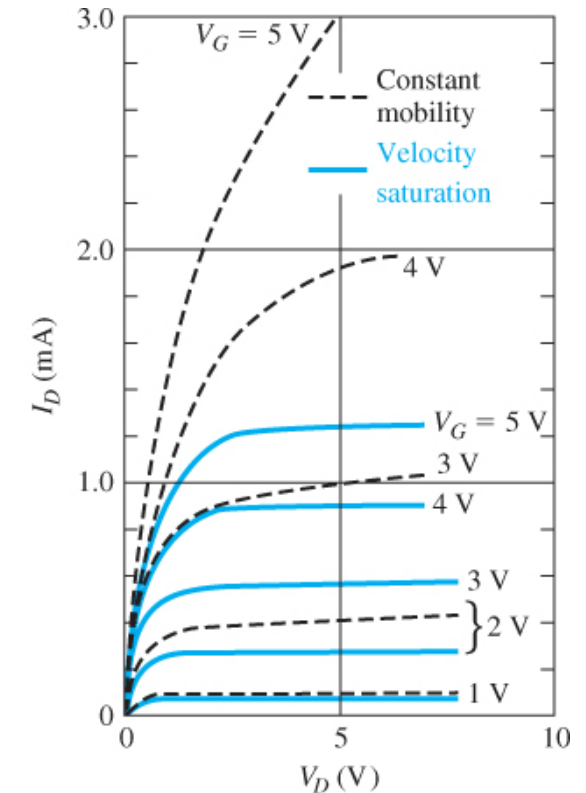


Figure 11.11 | Comparison of I_D versus V_D characteristics for constant mobility (dashed curves) and for field-dependent mobility and velocity saturation effects (solid curves).
(From Sze and Ng [22].)

MOSFET: Non-Ideal Effects



Subthreshold Current

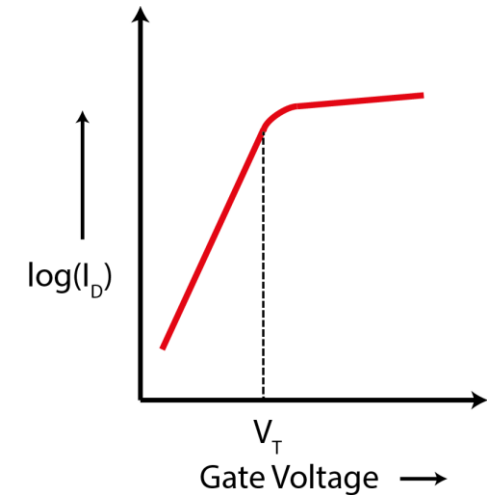
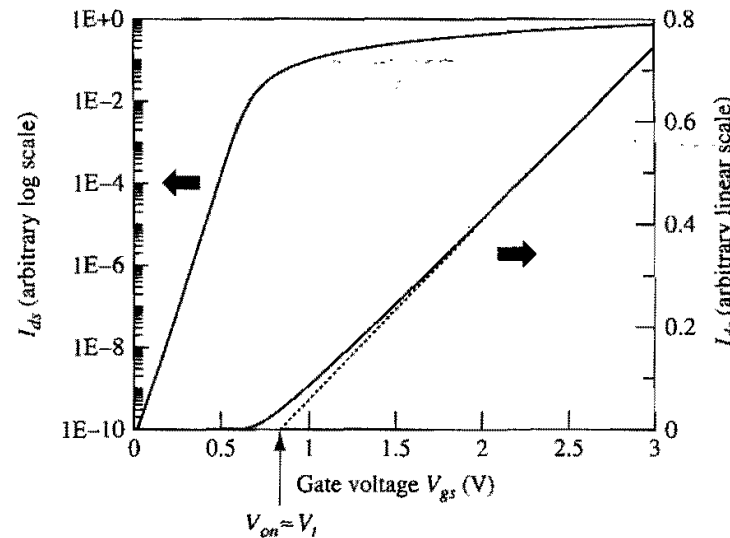
Subthreshold Current: Basics

For ideal MOSFET

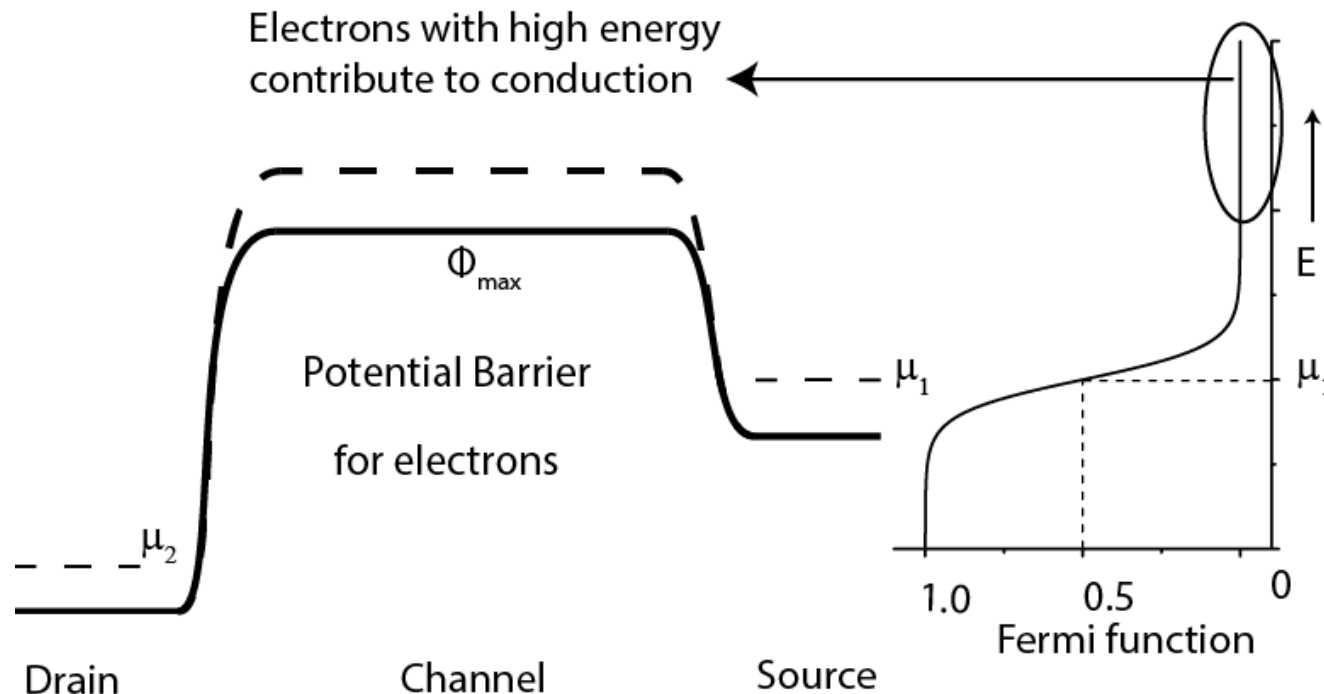
- We had assumed that the drain current is zero when $V_{GS} < V_T$

For real MOSFETs

- Small drain current flows even when $V_{GS} < V_T$
- This current is known as *subthreshold current*



Subthreshold Current: Reason



Source: S.Saurabh and M. J. Kumar, [Fundamentals of Tunnel Field Effect Transistors](#), CRC Press (Taylor & Francis), ISBN 9781498767132

- A potential barrier at the source stops electrons from moving from source to channel
- Some electron have energies higher than the potential barrier due to statistical distribution of energy (Fermi distribution function)
- Electrons in the “high-energy tail” are available for conduction in the subthreshold region [OFF-state] and contribute to subthreshold current

References

- D. A. Neaman, *"Semiconductor Physics and Devices"*